

OPTT Analysis

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```
#pull OPTH ticker data for all time since going public.
#date format: YYYY-MM-DD

optt <- tq_get("OPTT",
               get = "stock.prices",
               from = "2007-04-25", #went public on NASDAQ April 25th 2007
               to = "2026-06-21") #at time of download, update as needed
```

```
#preview data
```

```
head(optt)
```

```
# A tibble: 6 x 8
```

	symbol	date	open	high	low	close	volume	adjusted
	<chr>	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	OPTT	2007-04-25	4000	4000	3548	3568	15321	3568
2	OPTT	2007-04-26	3510	3590	3210	3336	6585	3336
3	OPTT	2007-04-27	3348	3436	3110	3110	1157	3110
4	OPTT	2007-04-30	3150	3196	2850	2868	1568	2868
5	OPTT	2007-05-01	2968	2968	2760	2860	1623	2860
6	OPTT	2007-05-02	2874	3250	2874	3184	1445	3184

```
#summary table
```

```
summary(optt)
```

symbol	date	open	high
Length:4819	Min. :2007-04-25	Min. : 0.12	Min. : 0.13
Class :character	1st Qu.:2012-02-02	1st Qu.: 1.04	1st Qu.: 1.09
Mode :character	Median :2016-11-15	Median : 44.60	Median : 47.00

	Mean	:2016-11-17	Mean	: 461.69	Mean	: 475.95	
	3rd Qu.:	2021-08-30	3rd Qu.:	644.00	3rd Qu.:	672.00	
	Max.	:2026-06-18	Max.	:4000.00	Max.	:4000.00	
low	close		volume		adjusted		
Min. :	0.12	Min. :	0.130	Min. :	0	Min. :	0.130
1st Qu.:	1.00	1st Qu.:	1.045	1st Qu.:	285	1st Qu.:	1.045
Median :	42.00	Median :	44.000	Median :	7710	Median :	44.000
Mean :	445.57	Mean :	459.070	Mean :	1974392	Mean :	459.070
3rd Qu.:	619.00	3rd Qu.:	639.000	3rd Qu.:	350200	3rd Qu.:	639.000
Max. :	3550.00	Max. :	3652.000	Max. :	616465100	Max. :	3652.000

```
#tidy datatable

#rename variables to be more
optt_tidy <- optt |>
  rename(open_price = open,
         high_price = high,
         low_price = low,
         close_price = close,
         Volume = volume,
         Adjusted = adjusted) |>

#break date into day, month, year columns
  separate(date, into = c("Year", "Month", "Day"), sep = "-", convert = TRUE) |>

#add month name column
  mutate(month_name = month.name[Month]) |>

#place month name column after month number
  relocate(month_name, .after = Month) |>

#create new variables
  mutate(
    #intra-day volatility
    idv = (high_price-low_price),
    #previous close price
    prev_close = lag(close_price),
    #absolute maximum upward movement
    max_up = (abs(high_price - prev_close)),
    #absolute maximum downward movement
    max_down = (abs(low_price - prev_close)),
    #true range:
```

```

#use pmax (parallel max) to take max for each day --> eliminates need for if/else + max(
true_range = pmax(idv, max_up, max_down, na.rm = TRUE),
#average true range:
#use rollmean (zoo package) to look at a sliding window
ATR = rollmean(true_range, k = 14, fill = NA, align = "right")
)

```

```
#preview tidy data
```

```
head(optt_tidy)
```

```
# A tibble: 6 x 17
```

```

symbol Year Month month_name Day open_price high_price low_price
<chr> <int> <int> <chr> <int> <dbl> <dbl> <dbl>
1 OPTT 2007 4 April 25 4000 4000 3548
2 OPTT 2007 4 April 26 3510 3590 3210
3 OPTT 2007 4 April 27 3348 3436 3110
4 OPTT 2007 4 April 30 3150 3196 2850
5 OPTT 2007 5 May 1 2968 2968 2760
6 OPTT 2007 5 May 2 2874 3250 2874
# i 9 more variables: close_price <dbl>, Volume <dbl>, Adjusted <dbl>,
# idv <dbl>, prev_close <dbl>, max_up <dbl>, max_down <dbl>,
# true_range <dbl>, ATR <dbl>

```

```
#view summary of tidy data
```

```
summary(optt_tidy)
```

symbol	Year	Month	month_name
Length:4819	Min. :2007	Min. : 1.000	Length:4819
Class :character	1st Qu.:2012	1st Qu.: 4.000	Class :character
Mode :character	Median :2016	Median : 7.000	Mode :character
	Mean :2016	Mean : 6.537	
	3rd Qu.:2021	3rd Qu.:10.000	
	Max. :2026	Max. :12.000	

Day	open_price	high_price	low_price
Min. : 1.00	Min. : 0.12	Min. : 0.13	Min. : 0.12
1st Qu.: 8.00	1st Qu.: 1.04	1st Qu.: 1.09	1st Qu.: 1.00
Median :16.00	Median : 44.60	Median : 47.00	Median : 42.00
Mean :15.72	Mean : 461.69	Mean : 475.95	Mean : 445.57

3rd Qu.:23.00	3rd Qu.: 644.00	3rd Qu.: 672.00	3rd Qu.: 619.00
Max. :31.00	Max. :4000.00	Max. :4000.00	Max. :3550.00

close_price	Volume	Adjusted	idv
Min. : 0.130	Min. : 0	Min. : 0.130	Min. : 0.00
1st Qu.: 1.045	1st Qu.: 285	1st Qu.: 1.045	1st Qu.: 0.09
Median : 44.000	Median : 7710	Median : 44.000	Median : 3.00
Mean : 459.070	Mean : 1974392	Mean : 459.070	Mean : 30.38
3rd Qu.: 639.000	3rd Qu.: 350200	3rd Qu.: 639.000	3rd Qu.: 34.00
Max. :3652.000	Max. :616465100	Max. :3652.000	Max. :658.00

prev_close	max_up	max_down	true_range
Min. : 0.13	Min. : 0.000	Min. : 0.00	Min. : 0.00
1st Qu.: 1.05	1st Qu.: 0.030	1st Qu.: 0.03	1st Qu.: 0.10
Median : 44.00	Median : 0.835	Median : 0.80	Median : 3.20
Mean : 459.17	Mean : 16.828	Mean : 15.33	Mean : 31.31
3rd Qu.: 639.50	3rd Qu.: 14.000	3rd Qu.: 14.00	3rd Qu.: 36.00
Max. :3652.00	Max. :694.000	Max. :476.00	Max. :694.00
NA's :1	NA's :1	NA's :1	

ATR

Min. : 0.0050
1st Qu.: 0.1036
Median : 4.1643
Mean : 31.0046
3rd Qu.: 42.6786
Max. :289.4286
NA's :13